Quasi-Newton Methods

Thus far we have worked with first order methods – those that use first derivative information to determine good descent directions

method	Pros	Cons
Gradient Desent	• simple implementation • best local descent • ok for small cond#	• poor global conv. rate • easily stalls
Conjugate Gradient	• superlinear convergence • minimal extra computation • minimal memory footprint	• requires restarts • restrictive line search ($c_2 < 1/2$)
Newton's Method	• quadratic convergence	• computationally expensive • large memory footprint

We next consider approximate second order methods that seek to gain good convergence without direct second order information. Quasi-Newton methods approximate the Hessian using curvature information derived from recent gradient computations.

Concept Example

$\min f(x) = \sin^2 x$ (near $x=0$)

Employ gradient descent with Armijo + Backtracking.
Start with $x_0 = 0.2$ and $c_1 = 0.001$ and $\alpha_0 = 1$

k	x_k	$f(x_k)$	$f'(x_k)$
0	+0.20000	+0.03947	+0.38942
1	−0.18942	+0.03545	−0.36984
2	+0.18042	+0.03200	+0.35306
3	−0.17264	0.02951	−0.33846

In this example, the performance is poor. The convergence rate appears to be linear: $\|x_{k+1}-x^*\|/\|x_k-x^*\| \approx$ const. Now consider a quasi-Newton approach. Begin with gradient descent so that $k=0$ and $k=1$ iterations are the same as before. Consider a quadratic model function near x_1.

$$m(p) = ap^2 + bp + c \quad (\text{hopefully with } a>0)$$

requiring: $m(0) = f(x_1)$, $m'(0) = f'(x_1)$, $m'(x_0-x_1) = f'(x_0)$

$$\Rightarrow \quad m(0) = c \overset{\text{set}}{=} 0.03546$$
$$m'(0) = b \overset{\text{set}}{=} -0.36984$$
$$m'(x_0-x_1) = 2a(0.38942)+b \overset{\text{set}}{=} 0.38942$$

The unique solution: $a = 0.97486$, $b = -0.36984$, $c = 0.03546$

The Newton step (using this model) is $p = -b/2a = 0.18975$.

$$x_2 = x_1 + p = 0.18942 + 0.18975 = 0.00033$$

$$f(x_2) = 1.1\times10^{-7} \qquad f'(x_2) = 6.6\times10^{-4}$$

The implementation in $\mathbb{R}^n$ is similar. When seeking p_{k+1}

$$m_{k+1}(p) = \tfrac{1}{2}p^T B_{k+1} p + \nabla f_{k+1}^T p + f_{k+1}$$

$$\nabla m_{k+1}(p) = B_{k+1}p + \nabla f_{k+1}$$

$$\nabla m_{k+1}(-\alpha_k p_k) \overset{\text{set}}{=} \nabla f_k \quad \#2$$

This last equation yields the so-called secant equation:

$$B_{k+1}(-\alpha_k p_k) + \nabla f_{k+1} = \nabla f_k$$
$$B_{k+1}(x_{k+1}-x_k) = \nabla f_{k+1} - \nabla f_k$$

$$\boxed{\begin{aligned} s_k &\equiv x_{k+1} - x_k \\ y_k &\equiv \nabla f_{k+1} - \nabla f_k \\ B_{k+1} s_k &= y_k \end{aligned}}$$

secant Equation

If we force B_{k+1} to be positive definite and that $\alpha_k p_k$ satisfies the curvature condition (second Wolfe condition) then the secant equation has a solution - but not unique. This means that we could have infinitely many update algs. The one that is considered the most beneficial (over decades of use) is the BFGS update.

let $H = B^{-1}$

$$H_{k+1} = \underset{J}{\operatorname{argmin}} \, \| J - H_k \|_F$$

$$\text{s.t.} \quad J = J^T$$

$$J y_k = s_k$$

with solution:

$$H_{k+1} = \left(I - \frac{s_k y_k^T}{y_k^T s_k} \right) H_k \left(I - \frac{y_k s_k^T}{y_k^T s_k} \right) + \frac{s_k s_k^T}{y_k^T s_k}$$

and (approximate) Newton Step

$$p_{k+1} = -B_{k+1}^{-1} \nabla f_{k+1} = -H_{k+1} \nabla f_{k+1}$$